

Kalman filter - for nonlinear systems

Sensors and Systems

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Outline

EKF and UKF

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

State estimation problem for nonlinear (NL) systems

Extended Kalman filter (EKF) for the NL DD problem

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State estimation problem for nonlinear (NL) systems

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

2

- ▶ Many real systems are more or less nonlinear.
- ▶ The important questions then are:
 - ▶ Is it possible to estimate states in NL systems?
 - ▶ Does *optimal* estimators exist?
 - ▶ Does *good* estimators exist?

The discrete-discrete (DD) NL estimation problem

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problem for
nonlinear (NL)
systems

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Kalman filter
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NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

Assume the system:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k) + w_k, \quad w_k \in \text{NID}(\underline{0}, Q_k), \\ y_k &= h(x_k, u_k) + v_k, \quad v_k \in \text{NID}(\underline{0}, R_k), \\ E(w(k)v(l)^T) &= \underline{0} \end{aligned}$$

Given measurements and initial values:

$$\begin{aligned} \mathcal{Y}_k &\triangleq y_0, y_1, y_2 \dots y_k, \quad u_0, u_1, u_2 \dots u_k \\ x_0 &\in N(\hat{x}_0, P_0) \end{aligned}$$

Find: $\hat{x}_k$ which minimized the mean square error

$$E((x_k - \hat{x}_k)^T M (x_k - \hat{x}_k)), \quad M \geq 0$$

The continuous-discrete (CD) NL estimation problem

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
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The state model is in continuous time i.e. a stochastic differential equation (SDE):

$$dx(t) = f(x(t), u(t))dt + d\omega(t) ,$$

$$\omega(t) \in W(Q(t)) ,$$

$$y(t_k) = h(x(t_k), u(t_k)) + v_k , v_k \in \text{NID}(\underline{0}, R_k)$$

Given measurements and initial values:

$$\mathcal{Y}_k \triangleq y(t_0), y(t_1), y(t_2) \dots y(t_k) , u(s), t_0 \leq s \leq t_k$$

$$x(t_0) \in N(\hat{x}(t_0), P(t_0))$$

Find: $\hat{x}(t)$ which minimized the mean square error

$$E((x(t) - \hat{x}(t))^T M (x(t) - \hat{x}(t))) , M \geq 0$$



Extended Kalman filter (EKF)

Basic idea

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estimation
problem for
nonlinear (NL)
systems

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NL DD
problem

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(UKF) for the
NL DD
problem

- ▶ The optimal estimator $E(x_k|\mathcal{Y}_k)$ can not normally be obtained.
- ▶ The KF can be used for a linearization of the NL system.
- ▶ This linearization can be made in different ways with fixed or variable linearization point.
- ▶ The best EKF is obtained by using the most resent state estimate as linearization point and using the non linear state and output functions where possible.

5

Linearization of f and h with respect to x is needed.

$$F_k(x) \triangleq \frac{\partial f(x)}{\partial x^T} ,$$

$$H_k(x) \triangleq \frac{\partial h(x)}{\partial x^T} .$$

- Notice $F_k \in R^{n \times n}$ and $H_k \in R^{p \times n}$ are matrices.
- The argument x to F_k and H_k should be the most recent estimate of x e.g. $\hat{x}_{k|k-1}$ or $\hat{x}_{k|k}$.

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

Initial conditions:

$$x_0 \in N(\hat{x}_{0|-1}, P_{0|-1})$$

Measurements update after receiving y_k and u_k :

$$\hat{y}_{k|k-1} = h(\hat{x}_{k|k-1}, u_k) ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k \triangleq P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} , \quad H_k = H_k(\hat{x}_{k|k-1})$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R_k K_k^T .$$

Time update from k to $k + 1$:

$$\hat{x}_{k+1|k} = f(\hat{x}_{k|k}, u_k) ,$$

$$P_{k+1|k} = F_k P_{k|k} F_k^T + Q_k , \quad F_k = F(\hat{x}_{k|k}) .$$

7

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Block diagram interpretation - mean value part (no direct feed though)

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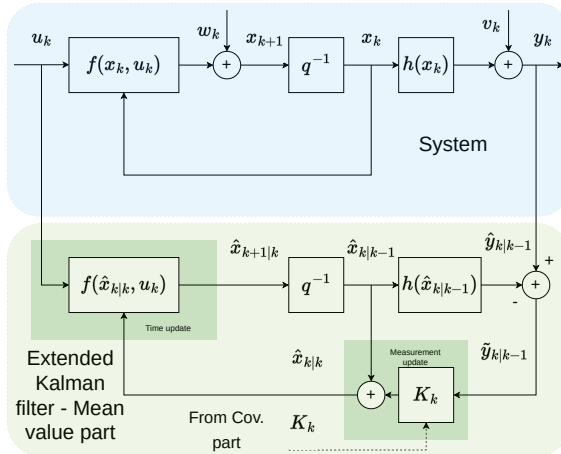
State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
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NL DD
problem

8



Block diagram interpretation - covariance value part

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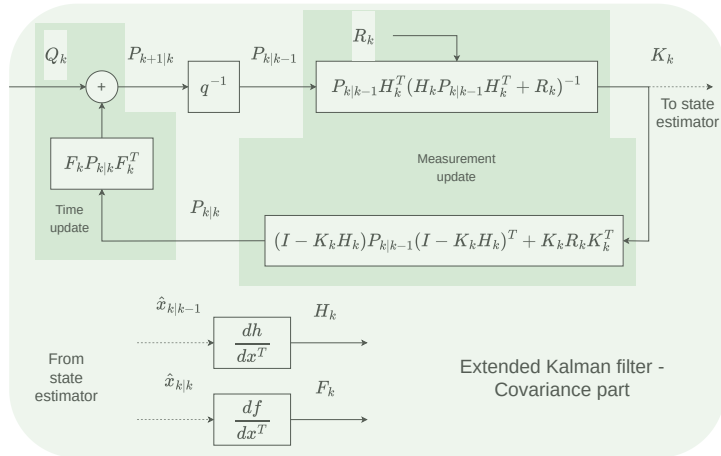
State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

9



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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

Important difficulties with the EKF

- ▶ The EKF is not optimal!
- ▶ The EKF is not guaranteed to be stable!

10

26



The unscented transform (UT)

Motivation

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problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

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transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

- ▶ In the derivation of the KF for linear systems the linear properties are extensively used and the final algorithm includes the system matrices.
- ▶ Linearization can be applied in the steps before linear system properties are used.

11

Example: time update from $\hat{x}_{k|k}, P_{k|k}$ to $\hat{x}_{k+1|k}, P_{k+1|k}$

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problem for
nonlinear (NL)
systems

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NL DD
problem

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Unscented
Kalman filter
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NL DD
problem

Non linear versus linear

In the linear case we have:

$$x_{k+1} = f(x_k, u_k) + w_k = \Phi_k x_k + \Gamma_k u_k + w_k \Rightarrow$$

$$\hat{x}_{k+1|k} = \mathbf{E}(x_{k+1} | \mathcal{Y}_k) \quad (\text{NL})$$

$$= \mathbf{E}(f(x_k, u_k) + w_k | \mathcal{Y}_k) \quad (\text{NL})$$

$$= \mathbf{E}(f(x_k, u_k) | \mathcal{Y}_k) \quad (\text{NL})$$

$$= \mathbf{E}(\Phi_k x_k + \Gamma_k u_k | \mathcal{Y}_k) \quad (\text{L})$$

$$= \Phi_k \hat{x}_{k|k} + \Gamma_k u_k \quad (\text{L})$$

12

26

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problem for
nonlinear (NL)
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problem

and

$$P_{k+1|k} = \text{Cov}(x_{k+1}|\mathcal{Y}_k) \quad (\text{NL})$$

$$= \text{Cov}(f(x_k, u_k) + w_k|\mathcal{Y}_k) \quad (\text{NL})$$

$$= \text{Cov}(f(x_k, u_k)|\mathcal{Y}_k) + Q_k \quad (\text{NL})$$

$$= \text{Cov}(\Phi_k x_k + \Gamma_k u_k|\mathcal{Y}_k) + Q_k \quad (\text{L})$$

$$= \Phi_k P_{k|k} \Phi_k + Q_k \quad (\text{L})$$

13

26

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
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NL DD
problem

The unscented
transform (UT)

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Kalman filter
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NL DD
problem

Notice also:

$$\begin{aligned} P_{k+1|k} &= \text{Cov}(x_{k+1}|\mathcal{Y}_k) \\ &= \text{E}((x_{k+1} - \text{E}(x_{k+1}|\mathcal{Y}_k))(x_{k+1} - \text{E}(x_{k+1}|\mathcal{Y}_k))^T|\mathcal{Y}_k) \\ &= \text{E}((x_{k+1} - \hat{x}_{k+1|k})(x_{k+1} - \hat{x}_{k+1|k})^T|\mathcal{Y}_k) \\ &= \text{E}(\tilde{x}_{k+1|k}\tilde{x}_{k+1|k}^T) \end{aligned}$$

$$\begin{aligned} \text{Cov}(x_k, y_k|\mathcal{Y}_{k-1}) &= \text{E}((x_k - \text{E}(x_k|\mathcal{Y}_{k-1}))(y_k - \text{E}(y_k|\mathcal{Y}_{k-1}))^T|\mathcal{Y}_{k-1}) \\ &= \text{E}((x_k - \hat{x}_{k|k-1})(y_k - \hat{y}_{k|k-1})^T|\mathcal{Y}_{k-1}) \\ &= \text{E}(\tilde{x}_{k|k-1}\tilde{y}_{k|k-1}^T) \end{aligned}$$

$$\begin{aligned} \text{Cov}(y_k|\mathcal{Y}_{k-1}) &= \text{E}((y_k - \text{E}(y_k|\mathcal{Y}_{k-1}))(y_k - \text{E}(y_k|\mathcal{Y}_{k-1}))^T|\mathcal{Y}_{k-1}) \\ &= \text{E}(\tilde{y}_{k|k-1}\tilde{y}_{k|k-1}^T) \end{aligned}$$

14

26

Kalman filter algorithm for linear DD problem

Reverse derivation to general expressions

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problem for
nonlinear (NL)
systems

Extended
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NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

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$$x_0 \in N(\hat{x}_{0|-1}, P_{0|-1})$$

Measurements update after receiving y_k and u_k :

$$\hat{y}_{k|k-1} = E(h(x_k, u_k) + v_k | \mathcal{Y}_{k-1}) ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k \triangleq \text{Cov}(x_k, y_k | \mathcal{Y}_{k-1}) \text{Cov}(y_k | \mathcal{Y}_{k-1})^{-1} ,$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = P_{k|k-1} - K_k \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1}) .$$

Time update from k to $k + 1$:

$$\hat{x}_{k+1|k} = E(f(x_k, u_k) + w_k | \mathcal{Y}_k) ,$$

$$P_{k+1|k} = \text{Cov}(f(x_k, u_k) + w_k | \mathcal{Y}_k)$$

15

26

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
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problem

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transform (UT)

Unscented
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NL DD
problem

The generic probabilistic problem

Calculate/approximate the second order statistics of $x, y = f(x)$ given the second order statistics of x and the function f :

$$y = f(x), \mu_x = E(x), C_x = \text{Cov}(x), x \in R^n, y \in R^p$$

For non-linear relations there is no exact solution in general.

The linearization solution

The EKF can be interpreted as using the linearization approach:

$$y = f(\mu_x) + \nabla f(\mu_x)(x - \mu_x) \Rightarrow$$

$$\mu_y = f(\mu_x), C_{yx} = \nabla f(\mu_x)C_x,$$

$$C_y = \nabla f(\mu_x)C_x \nabla f(\mu_x)^T.$$

16

26

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
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Unscented
Kalman filter
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NL DD
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- ▶ A general approach to approximation in probability problems is *Monte Carlo (MC) simulations*.
- ▶ In MC a number of random samples of x is generated, corresponding $y = f(x)$ are calculated and the empirical estimates of means and covariances are used as approximations.

$$y_i = f(x_i), \quad i = 1, \dots, N, \quad \hat{\mu}_y = \bar{y} = \frac{1}{N} \sum_{i=1}^N f(x_i),$$

$$\hat{C}_{yx} = \frac{1}{N} \sum_{i=1}^N (f(x_i) - \hat{\mu}_y)(x_i - \bar{x})^T, \quad \hat{C}_y = \frac{1}{N} \sum_{i=1}^N (f(x_i) - \hat{\mu}_y)(f(x_i) - \hat{\mu}_y)^T$$

- ▶ The UT method is similar to MC except the realizations of the random variable are not “random” but constructed and the number used is very limited compared to MC.

17

26

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

18

$\alpha = 1$, $\kappa = 2$, $\beta = 0$, (Default parameters gives $\lambda = 2$)

$\lambda = \alpha^2(n + \kappa) - n$, $k = \sqrt{n + \lambda}$, $n = \dim(x)$,

$u_i \triangleq$ eigenvector i for C_x , $l_i \triangleq$ eigenvalue i for C_x ,

$$x_i = \begin{cases} \mu_x & , i = 0 \\ \mu_x + k u_i \sqrt{l_i} & , 1 \leq i \leq n \\ \mu_x - k u_i \sqrt{l_i} & , n + 1 \leq i \leq 2n \end{cases}$$

$$w_i^t = \begin{cases} \frac{\lambda}{n + \lambda} & , i = 0 , t = m \text{ (for mean)} \\ \frac{\lambda}{n + \lambda} + 1 - \alpha^2 + \beta & , i = 0 , t = c \text{ (for covarians)} \\ \frac{1}{2(n + \lambda)} & , 1 \leq i \leq 2n \end{cases}$$

$$\hat{\mu}_y = \sum_{i=0}^{2n} w_i^m f(x_i) , \hat{C}_{y_x} = \sum_{i=0}^{2n} w_i^c (f(x_i) - \hat{\mu}_y)(x_i - \hat{\mu}_x)^T ,$$

$$\hat{C}_y = \sum_{i=0}^{2n} w_i^c (f(x_i) - \hat{\mu}_y)(f(x_i) - \hat{\mu}_y)^T .$$

Important knowledge about the UT

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

- ▶ There are many alternative ways to do this. They are some times called *sampling* methods.
- ▶ The UT gives exact results if f is affine $f(x) = Ax + b$ which it also should.
- ▶ For NL f it is still an approximation but often gives good result.
- ▶ Especially it is generally superior to the linearization method and the MC method.

19

26

Example

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problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

20

In order to have a test where the correct result is known $y = x^T x$ is used with x Gaussian and $E(x) = [1; 1]$ and $Cov(x) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ (using Matlab notation).

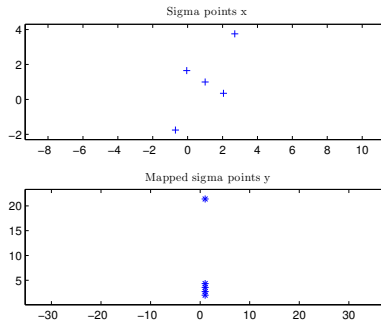


Figure: Sigma points and images for the UT.

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

21

Unscented
Kalman filter
(UKF) for the
NL DD
problem

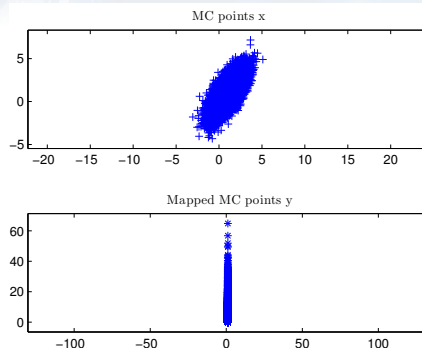


Figure: Random (sigma) points and images for the MC based method.

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

22

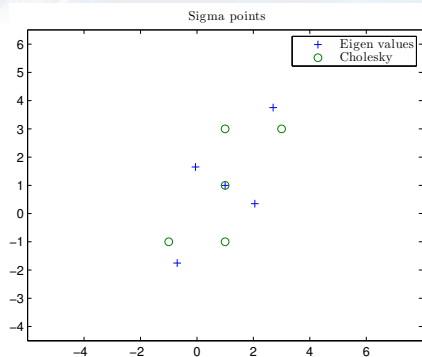


Figure: Comparison of sigma points based on eigenvalues and the Cholesky method.

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

23

Method	My	Cy	Cyx	
UTSCEst	5.000e+0	3.900e+1	4.000e+0	6.000e+0
UTJMEst	5.000e+0	3.100e+1	4.000e+0	6.000e+0
MCEst	5.038e+0	3.385e+1	3.988e+0	6.028e+0
THTK	5.000e+0	3.400e+1	4.000e+0	6.000e+0
UTSCRelE	0	1.471e-1	0	0
UTJMRelE	0	-8.824e-2	0	0
MCRelE	7.663e-3	-4.284e-3	-3.090e-3	4.670e-3

Table: Performance for the above method, the Cholesky based, the MC and the theoretical. The rows marked RelE are the errors relative to the correct results.

26



Unscented Kalman filter

Definition

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

Assume the NL map f with input x and output y , the UT can then be represented by the function U below:

$$\begin{bmatrix} E(y) \\ \text{Cov}(y, x) \\ \text{Cov}(y) \end{bmatrix} = U(f, E(x), \text{Cov}(x))$$

If the mean and covariance input is conditioned on say z then the output is also conditioned on z i.e.:

$$\begin{bmatrix} E(y|z) \\ \text{Cov}(y, x|z) \\ \text{Cov}(y|z) \end{bmatrix} = U(f, E(x|z), \text{Cov}(x|z))$$

24

26

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

Initial conditions:

$$x_0 \in N(\hat{x}_{0|-1}, P_{0|-1})$$

Measurements update after receiving y_k and u_k :

$$\begin{bmatrix} \hat{y}_{k|k-1} \\ \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1}) \\ C_k^h \end{bmatrix} = U(h, \hat{x}_{k|k-1}, P_{k|k-1}) ,$$

$$\text{Cov}(y_k | \mathcal{Y}_{k-1}) = C_k^h + R_k ,$$

$$\text{Cov}(x_k, y_k | \mathcal{Y}_{k-1}) = \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1})^T ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k = \text{Cov}(x_k, y_k | \mathcal{Y}_{k-1}) \text{Cov}(y_k | \mathcal{Y}_{k-1})^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = P_{k|k-1} - K_k \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1}) .$$

25

26

EKF and UKF

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
NL DD
problem

Time update from k to $k + 1$:

$$\begin{bmatrix} \hat{x}_{k+1|k} \\ X \\ C_k^f \end{bmatrix} = U(f, \hat{x}_{k|k}, P_{k|k}) ,$$

$$P_{k+1|k} = \text{Cov}(x_{k+1} | \mathcal{Y}_k) = C_k^f + Q_k .$$

26

26